



MME 321 Lab 10: Transfer function + State System

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1.0 Transfer Function and State-Space Modeling in MATLAB

This lab applied Laplace transforms to convert linear differential equations into algebraic forms in the sss-domain and to construct equivalent transfer function and state-space models in MATLAB. The following examples are discussed in this lab.

1. $y'' + 5y' + 6y = 2u$
2. $2y'' + 7y' + 3y = 4u' + u$
3. $y''' + 6y'' + 11y' + 6y = u$
4. $3y''' + 2y'' + 5y' + y = 2u$
5. $y'''' + 4y''' + 8y'' + 6y' + 2y = u$
6. $2y'''' + 5y''' + 3y'' + 7y' + 4y = 3u'' + 2u$
7. $y'''' + 5y''' + 9y'' + 7y' + 3y + y = u$
8. $4y''' + 12y'' + 9y' + 2y = 5u' + 3u$

2.0 Laplace Transform Review

The Laplace transform is used to convert time-domain functions into functions of frequency domain. The Laplace transform allows differential equations to be written as algebraic. This is useful for finding transfer functions and predicting dynamic responses. Zero initial conditions are assumed for this method.

The main derivative properties used in this lab were:

$$\begin{aligned}L\{dy/dt\} &= sY(s) - y(0) \\L\{d^2y/dt^2\} &= s^2Y(s) - sy(0) - y'(0) \\L\{d^ny/dt^n\} &= s^nY(s) - s^{n-1}y(0) - \dots - y^{(n-1)}(0)\end{aligned}$$

Under zero initial conditions:

$$\begin{aligned}L\{y'\} &= sY(s) \\L\{y''\} &= s^2Y(s) \\L\{y'''\} &= s^3Y(s)\end{aligned}$$

common transform pairs used in this lab:

$$\begin{aligned}L\{u(t)\} &= 1/s \\L\{t\} &= 1/s^2 \\L\{t^2\} &= 2/s^3 \\L\{e^{-at}\} &= 1/s + a \\L\{\sin(\omega t)\} &= \omega/s^2 + \omega^2 \\L\{\cos(\omega t)\} &= s/s^2 + \omega^2\end{aligned}$$

3.0 Transfer Function Derivation

A transfer function is defined as the ratio of the output function to input function in laplace domain assuming only zero initial conditions:

$$G(s) = Y(s)/U(s).$$

Example 1

$$y'' + 5y' + 6y = 2u$$

$$s^2Y(s) + 5sY(s) + 6Y(s) = 2U(s)$$

$$Y(s)(s^2 + 5s + 6) = 2U(s)$$

$$G(s) = Y(s)$$

$$U(s) = 2/(s^2 + 5s + 6)$$

Example 2

$$2y'' + 7y' + 3y = 4u' + u$$

$$2s^2Y(s) + 7sY(s) + 3Y(s) = 4sU(s) + U(s)$$

$$Y(s)(2s^2 + 7s + 3) = U(s)(4s + 1)$$

$$G(s) = Y(s)/U(s) = (4s + 1)/(2s^2 + 7s + 3)$$

Example 3

$$y''' + 6y'' + 11y' + 6y = u$$

$$s^3Y(s) + 6s^2Y(s) + 11sY(s) + 6Y(s) = U(s)$$

$$Y(s)(s^3 + 6s^2 + 11s + 6) = U(s)$$

$$G(s) = 1/s^3 + 6s^2 + 11s + 6$$

Example 4

$$3y''' + 2y'' + 5y' + y = 2u$$

$$3s^3Y(s) + 2s^2Y(s) + 5sY(s) + Y(s) = 2U(s)$$

$$Y(s)(3s^3 + 2s^2 + 5s + 1) = 2U(s)$$

$$G(s) = 2/(3s^3 + 2s^2 + 5s + 1)$$

Example 5

$$y'''' + 4y''' + 8y'' + 6y' + 2y = u$$

$$s^4Y(s) + 4s^3Y(s) + 8s^2Y(s) + 6sY(s) + 2Y(s) = U(s)$$

$$Y(s)(s^4 + 4s^3 + 8s^2 + 6s + 2) = U(s)$$

$$G(s) = 1/s^4 + 4s^3 + 8s^2 + 6s + 2$$

Example 6

$$2y'''' + 5y''' + 3y'' + 7y' + 4y = 3u'' + 2u$$

$$2s^4Y(s) + 5s^3Y(s) + 3s^2Y(s) + 7sY(s) + 4Y(s) = 3s^2U(s) + 2U(s)$$

$$Y(s)(2s^4 + 5s^3 + 3s^2 + 7s + 4) = U(s)(3s^2 + 2)$$

$$G(s) = (3s^2 + 2)/(2s^4 + 5s^3 + 3s^2 + 7s + 4)$$

Example 7

$$y''''' + 5y'''' + 9y''' + 7y'' + 3y' + y = u$$

$$s^5Y(s) + 5s^4Y(s) + 9s^3Y(s) + 7s^2Y(s) + 3sY(s) + Y(s) = U(s)$$

$$Y(s)(s^5 + 5s^4 + 9s^3 + 7s^2 + 3s + 1) = U(s)$$

$$G(s) = 1/(s^5 + 5s^4 + 9s^3 + 7s^2 + 3s + 1)$$

Example 8

$$4y''' + 12y'' + 9y' + 2y = 5u' + 3u$$

$$4s^3Y(s) + 12s^2Y(s) + 9sY(s) + 2Y(s) = 5sU(s) + 3U(s)$$

$$Y(s)(4s^3 + 12s^2 + 9s + 2) = U(s)(5s + 3)$$

$$G(s) = (5s + 3)/(4s^3 + 12s^2 + 9s + 2)$$

4.0 MATLAB TF Code

% Example 1

% $y'' + 5y' + 6y = 2u$

U1 = [2];

Y1 = [1 5 6];

G1 = tf(U1, Y1);

figure;

step(G1);

grid on;

title('Step Response for Example 1');

xlabel('Time (s)');

ylabel(' y(t)');

% Example 2

% $2y'' + 7y' + 3y = 4u' + u$

U2 = [4 1];

Y2 = [2 7 3];

G2 = tf(U2, Y2);

figure;

step(G2);

grid on;

title('Step Response for Example 2');

xlabel('Time (s)');

ylabel(' y(t)');

% Example 3

% $y''' + 6y'' + 11y' + 6y = u$

U3 = [1];

```
Y3 = [1 6 11 6];  
G3 = tf(U3, Y3);
```

```
figure;  
step(G3);  
grid on;  
title('Step Response for Example 3');  
xlabel('Time (s)');  
ylabel(' y(t)');
```

```
% Example 4  
%  $3y''' + 2y'' + 5y' + y = 2u$ 
```

```
U4 = [2];  
Y4 = [3 2 5 1];
```

```
G4 = tf(U4, Y4);
```

```
figure;  
step(G4);  
grid on;  
title('Step Response for Example 4');  
xlabel('Time (s)');  
ylabel(' y(t)');
```

```
% Example 5  
%  $y'''' + 4y''' + 8y'' + 6y' + 2y = u$ 
```

```
U5 = [1];  
Y5 = [1 4 8 6 2];  
G5 = tf(U5, Y5);
```

```
figure;  
step(G5);  
grid on;  
title('Step Response for Example 5');  
xlabel('Time (s)');  
ylabel(' y(t)');
```

```
% Example 6
```

```
%  $2y'''' + 5y''' + 3y'' + 7y' + 4y = 3u'' + 2u$ 
```

```
U6 = [3 0 2];
```

```
Y6 = [2 5 3 7 4];
```

```
G6 = tf(U6, Y6);
```

```
figure;
```

```
step(G6);
```

```
grid on;
```

```
title('Step Response for Example 6');
```

```
xlabel('Time (s)');
```

```
ylabel(' y(t)');
```

```
% Example 7
```

```
%  $y'''' + 5y''' + 9y'' + 7y' + y = u$ 
```

```
U7 = [1];
```

```
Y7 = [1 5 9 7 3 1];
```

```
G7 = tf(U7, Y7);
```

```
figure;
```

```
step(G7);
```

```
grid on;
```

```
title('Step Response for Example 7');
```

```
xlabel('Time (s)');
```

```
ylabel(' y(t)');
```

```
% Example 8
```

```
%  $4y'''' + 12y''' + 9y'' + 2y' = 5u' + 3u$ 
```

```
U8 = [5 3];
```

```
Y8 = [4 12 9 2];
```

```
G8 = tf(U8, Y8);
```

```
figure;
```

```
step(G8);
```

```
grid on;
```

```
title('Step Response for Example 8');
```

```
xlabel('Time (s)');
```

ylabel(' y(t)');

5.0 State-Space Formulation

State-space modeling represents a dynamic system as a set of first-order differential equations that can be used when there are non-zero initial values. The system is written as:

$$\dot{x} = Ax + Bu$$

$$y = Cx + Du$$

Controllable canonical form says that A is the state variable coefficients, B is the input variable coefficients.

Example 1

$$y'' + 5y' + 6y = 2u$$

Choose the states as

$$x_1 = y, x_2 = y'$$

$$x_1' = x_2$$

$$x_2' = -6x_1 - 5x_2 + 2u$$

Matrices:

$$A = \begin{bmatrix} 0 & 1 \\ -6 & -5 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

$$D = 0$$

Example 2

$$2y'' + 7y' + 3y = 4u' + u$$

$$G(s) = Y(s)/U(s) = (4s + 1)/(2s^2 + 7s + 3)$$

Controllable canonical form

$$A = \begin{bmatrix} 0 & 1 \\ -3/2 & -7/2 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$C = [1 \ 2]$$

$$D = 0$$

Example 3

$$y''' + 6y'' + 11y' + 6y = u$$

Choose the states

$$x_1 = y, x_2 = y', x_3 = y''$$

$$x_1' = x_2$$

$$x_2' = x_3$$

$$x_3' = -6x_1 - 11x_2 - 6x_3 + u$$

Matrix

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}'$$

$$C = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$$

$$D = 0$$

Example 4

$$3y''' + 2y'' + 5y' + y = 2u$$

$$G(s) = 2/(3s^3 + 2s^2 + 5s + 1)$$

controllable canonical form

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1/3 & -5/3 & -2/3 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}'$$

$$C = \begin{bmatrix} 2/3 & 0 & 0 \end{bmatrix}$$

$$D = 0$$

Example 5

$$y'''' + 4y''' + 8y'' + 6y' + 2y = u$$

Choose states

$$x_1 = y, x_2 = y', x_3 = y'', x_4 = y'''$$

$$\begin{aligned}x_1' &= x_2, \quad x_2' = x_3, \quad x_3' = x_4 \\x_4' &= -2x_1 - 6x_2 - 8x_3 - 4x_4 + u\end{aligned}$$

Matrix

$$\begin{aligned}A &= [0 \ 1 \ 0 \ 0; 0 \ 0 \ 1 \ 0; 0 \ 0 \ 0 \ 1; -2 \ -6 \ -8 \ -4] \\B &= [0 \ 0 \ 0 \ 1]' \\C &= [1 \ 0 \ 0 \ 0] \\D &= 0\end{aligned}$$

Example 6

$$2y'''' + 5y''' + 3y'' + 7y' + 4y = 3u'' + 2u$$

The transfer function is

$$G(s) = (3s^2 + 2)/(2s^4 + 5s^3 + 3s^2 + 7s + 4)$$

controllable canonical form

$$\begin{aligned}A &= [0 \ 1 \ 0 \ 0; 0 \ 0 \ 1 \ 0; 0 \ 0 \ 0 \ 1; -2 \ -7/2 \ -3/2 \ -5] \\B &= [0 \ 0 \ 0 \ 1]' \\C &= [1 \ 0 \ 3/2 \ 0] \\D &= 0\end{aligned}$$

Example 7

Differential equation:

$$y'''' + 5y''' + 9y'' + 7y' + 3y + y = u$$

Choose states

$$\begin{aligned}x_1 &= y \\x_2 &= y' \\x_3 &= y'' \\x_4 &= y''' \\x_5 &= y''''\end{aligned}$$

Then

$$x_1' = x_2$$

$$\begin{aligned}x_2' &= x_3 \\x_3' &= x_4 \\x_4' &= x_5 \\x_5' &= -x_1 - 3x_2 - 7x_3 - 9x_4 - 5x_5 + u\end{aligned}$$

matrix

$$\begin{aligned}A &= [0 \ 1 \ 0 \ 0 \ 0; 0 \ 0 \ 1 \ 0 \ 0; 0 \ 0 \ 0 \ 1 \ 0; 0 \ 0 \ 0 \ 0 \ 1; -1 \ -3 \ -7 \ -9 \ -5] \\B &= [0 \ 0 \ 0 \ 0 \ 1]' \\C &= [1 \ 0 \ 0 \ 0 \ 0] \\D &= 0\end{aligned}$$

Example 8

$$4y''' + 12y'' + 9y' + 2y = 5u' + 3u$$

transfer function

$$G(s) = (5s + 3)/(4s^3 + 12s^2 + 9s + 2)$$

controllable canonical form

$$\begin{aligned}A &= [0 \ 1 \ 0; 0 \ 0 \ 1; -1/2 \ -9/4 \ -3] \\B &= [0 \ 0 \ 1]' \\C &= [3/4 \ 5/4 \ 0] \\D &= 0\end{aligned}$$

6.0 MATLAB State-Space Code

% Example 1

```
A1 = [0 1; -6 -5];
B1 = [0; 2];
C1 = [1 0];
D1 = 0;
Ex1 = ss(A1,B1,C1,D1);
```

```
figure;
step( Ex1);
grid on;
title('State-Space Step Response - Example 1');
xlabel('Time (s)');
```

```
ylabel(' y(t)');
```

```
% Example 2
```

```
A2 = [0 1; -3/2 -7/2];
```

```
B2 = [0; 1];
```

```
C2 = [1/2 2];
```

```
D2 = 0;
```

```
Ex2 = ss(A2,B2,C2,D2);
```

```
figure;
```

```
step( Ex2);
```

```
grid on;
```

```
title('State-Space Step Response - Example 2');
```

```
xlabel('Time (s)');
```

```
ylabel(' y(t)');
```

```
% Example 3
```

```
A3 = [0 1 0; 0 0 1; -6 -11 -6];
```

```
B3 = [0; 0; 1];
```

```
C3 = [1 0 0];
```

```
D3 = 0;
```

```
Ex3 = ss(A3,B3,C3,D3);
```

```
figure;
```

```
step( Ex3);
```

```
grid on;
```

```
title('State-Space Step Response - Example 3');
```

```
xlabel('Time (s)');
```

```
ylabel(' y(t)');
```

```
% Example 4
```

```
A4 = [0 1 0; 0 0 1; -1/3 -5/3 -2/3];
```

```
B4 = [0; 0; 1];
```

```
C4 = [2/3 0 0];
```

```
D4 = 0;
```

```
Ex4 = ss(A4,B4,C4,D4);
```

```
figure;
```

```
step( Ex4);
```

```
grid on;  
title('State-Space Step Response - Example 4');  
xlabel('Time (s)');  
ylabel(' y(t)');
```

```
% Example 5  
A5 = [0 1 0 0; 0 0 1 0; 0 0 0 1; -2 -6 -8 -4];  
B5 = [0; 0; 0; 1];  
C5 = [1 0 0 0];  
D5 = 0;  
Ex5 = ss(A5,B5,C5,D5);
```

```
figure;  
step( Ex5);  
grid on;  
title('State-Space Step Response - Example 5');  
xlabel('Time (s)');  
ylabel(' y(t)');
```

```
% Example 6  
A6 = [0 1 0 0; 0 0 1 0; 0 0 0 1; -2 -7/2 -3/2 -5/2];  
B6 = [0; 0; 0; 1];  
C6 = [1 0 3/2 0];  
D6 = 0;  
Ex6 = ss(A6,B6,C6,D6);
```

```
figure;  
step( Ex6);  
grid on;  
title('State-Space Step Response - Example 6');  
xlabel('Time (s)');  
ylabel(' y(t)');
```

```
% Example 7  
A7 = [0 1 0 0 0; 0 0 1 0 0; 0 0 0 1 0; 0 0 0 0 1; -1 -3 -7 -9 -5];  
B7 = [0; 0; 0; 0; 1];  
C7 = [1 0 0 0 0];  
D7 = 0;  
Ex7 = ss(A7,B7,C7,D7);
```

```
figure;
step( Ex7);
grid on;
title('State-Space Step Response - Example 7');
xlabel('Time (s)');
ylabel(' y(t)');
```

```
% Example 8
A8 = [0 1 0; 0 0 1; -1/2 -9/4 -3];
B8 = [0; 0; 1];
C8 = [3/4 5/4 0];
D8 = 0;
Ex8 = ss(A8,B8,C8,D8);
```

```
figure;
step( Ex8);
grid on;
title('State-Space Step Response - Example 8');
xlabel('Time (s)');
ylabel(' y(t)');
```

7.0 Comparison of Responses

Step responses from transfer function and state-space models were compared for all eight systems. The results showed complete overlap in every case, confirming that both formulations represent identical system dynamics when derived correctly. Key response characteristics—rise time, settling time, transient shape, and steady-state value—matched precisely.

Minor visual discrepancies, when present, were attributed to numerical precision or plotting resolution rather than modeling errors. Since transfer functions capture the input-output relationship and state-space models describe internal dynamics, their equivalence demonstrates consistency between external and internal system representations.

Any mismatch between responses would indicate errors such as incorrect coefficient extraction, sign inconsistencies, or improper state definitions. No such issues were observed.

These results validate that transfer function and state-space approaches are mathematically equivalent and interchangeable for linear time-invariant systems under zero initial conditions.

8.0 Discussion and Conclusion

Laplace transforms simplify differential equations into algebraic expressions, enabling efficient system modeling and analysis. Transfer functions provide a compact input-output description, useful for analyzing poles, zeros, and stability. However, they do not explicitly describe internal system behavior.

State-space models offer a complete system representation, capturing internal state dynamics and supporting multi-input, multi-output systems. They are better suited for numerical simulation and modern control design.

The equivalence of both methods was verified through identical step responses. This confirms that both representations describe the same physical system in different mathematical forms.

For higher-order and complex systems, state-space modeling provides greater flexibility, while transfer functions remain valuable for analytical insight.